

ROAN HOWARD

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EDUCATION

Stanford University, Stanford, CA

June 2027

BS in Computer Science, specializing in Artificial Intelligence; minor in Art Practice

Relevant Coursework: Linear Algebra and Multivariable Calculus (MATH51); Mathematical Foundations of Computing (CS103); Mechanics (PHYSICS41); Statistical Methods (STATS60); Probability for Computer Scientists (CS109); Programming Methodology (CS106A); Programming Abstractions (CS106B); Computer Organization and Systems (CS107); Information Science and Engineering (ENGR76); Design and Analysis of Algorithms (CS161); Natural Language Processing & Large Language Models (CS124); Matrix Methods (ENGR108); Continuous Mathematical Methods with an Emphasis on Machine Learning (CS205L); Machine Learning (CS229)

Student Organizations: Stanford Alaska Native Student Association; The American Indian Science and Engineering Society, Stanford Chess Club; Stanford Orientation Volunteer (Logistics for 1,500+ students); Stanford Tree Hacks (acceptance rate ~7%); Kappa Sigma Fraternity, Intramural Basketball

Inglemoor High School, Kenmore, WA

June 2023

The International Baccalaureate Diploma Program; Chess Club President; Economics Club; Inglemoor High School Honors Society; Varsity Athlete: Soccer, Cross Country, Crew; Orientation Leader; Eastside Native American Program Coordinator

RECENT PROJECTS (more on website)

Hubbard Brook Agent, <https://hbf-agent.vercel.app>

April 2026

- Built and deployed a full-stack AI agent featuring an animated Abenaki figure that reacts in real time to live Hubbard Brook sensor data, submitted to the 2026 Hubbard Brook Data Jam Art category
- Engineered a multi-tool Claude Sonnet backend using FastAPI, Anthropic SDK, Next.js, enabling natural language queries over 60+ years of environmental records with live weather retrieval and historical trend analysis

Enterprise Support Agent

March 2026

- Built an Agentic RAG chatbot using Claude Sonnet Anthropic SDK, FAISS vector search, and sentence-transformers for semantic retrieval with citation-grounded responses
- Implemented native tool use with multi-turn memory, FastAPI backend, and a Tailwind dashboard UI

Movie Service Agent

February 2026

- Engineered a multi-tool LLM agent leveraging DSPy ReAct, Mixtral-8x7B, and cosine-similarity collaborative filtering for real-time movie recommendations and ticket booking
- Augmented agent with Agentic RAG capabilities via SerpAPI retrieval and Mem0 semantic memory using Qdrant vector store and Alibaba-NLP dense embeddings on Together AI

Robot Data Labeler

February 2026

- In 12 hours at the Stanford annual hackathon, built a video-reasoning robotics referee designed around the Stanford IPRL Lab dataset using Python, FastAPI, Next.js, and Dockerized local deployment
- Engineered automated video auditing using Cosmos Reason API, OpenAI vision models, and structured JSON evaluation schemas to assess robot demonstration videos against goal and rule constraints

RECENT WORK EXPERIENCE

Agentic AI Research Group, *Researcher*, Stanford, CA

March 2026 - Present

- Under PhD student Alexander Johansen, training an MLP to replace a hardcoded sparsity parameter with query-adaptive prediction inside a Koopman operator-based multi-agent LLM retrieval system
- Authoring a research paper benchmarking learned λ granularity control against fixed-parameter baselines across multi-hop and single-hop QA tasks in a 3-tier specialist agent architecture

New England Seafood International, *Learning and Development Intern*, London, UK

June 2025 - August 2025

- Revamped learning and development strategy for 600+ employees
- Deployed a Python script to automate Dayforce course uploads, saving 100+ hours of manual administrative time

Van Ness Feldman, *Legal Assistant*, Seattle, WA

June 2024 - August 2024

- Contributed to the Ninth Circuit's Wild Fish Conservancy ruling, securing economic stability for 15,000+ Southeast Alaskans
- Researched over \$100K in potential grant opportunities for Alaska Native corporations

SKILLS AND INTERESTS

- *Programming Languages:* Python, Julia, C++, C, Assembly, LaTeX, HTML, CSS, R, SQL, MATLAB
- *Libraries:* NumPy, Pandas, Matplotlib, scikit-learn, SciPy, TensorFlow, PyTorch, Keras and more.
- *Tools:* Jupyter, Hugging Face, Nebius, Supabase, Vercel, GitHub, pgAdmin, Figma, DaVinci Resolve, Excel
- *Hobbies:* Ran 2025 Vancouver Marathon, author of 300+ film reviews, backpacked 15+ countries, learning German, sculptor